

ML Specification Report — cat-in-the-dat

Categorical Feature Encoding Challenge · our from-scratch agent (sparse-OHE linear pool + tree-search v3)

Figures grounded in the competition's `STATUS.md`, `config.yaml`, `dossier.json`, `experiments.json`, `experiments_tree_v3.json`, and `headless_run.log`, with leaderboard and cross-agent figures from the study's frozen three-way score table.

Overview

The task is binary classification from **23 exclusively categorical features** — the 2019 “cat-in-the-dat” challenge, explicitly designed as an encoding benchmark (metric: ROC-AUC, maximize). We solve it with a from-scratch pipeline — sparse one-hot + regularized logistic regression as the primary learner family, with LightGBM target-encoded and native-categorical members, refined through a linear protocol and a full 60-node `harness_v3` tree search. Our champion, tree-search node #55 (a kitchen-sink mega-blend of all 38 solos, 17 nonzero weights, LR-family-dominated), scores **CV OOF AUC 0.803272** (honest leave-fold-out weight re-fit 0.803231). On the real leaderboard it reaches **Public 0.80801 / Private 0.80241 — rank 362/1341, percentile 73.1** — winning the three-way benchmark on both local CV and the leaderboard: AIDE lands a razor-thin second (Private 0.80217), and the NVIDIA reproduce-agent trails far behind (Private 0.77084).

Why it matters. Categorical encoding is the unavoidable first decision of virtually every industrial tabular problem — IDs, product codes, hashed keys. This competition isolates that decision from everything else, so the answer it yields (when a sparse linear model beats GBDT, and how to target-encode without leaking the label) transfers directly to real-world pipelines.

Data

Purpose of Data. Predict a binary `target` from categorical columns only — an i.i.d. tabular binary-classification benchmark with no numeric features at all. **Data Format** is clean **tabular CSV**. **Data Volume** is **300,000 training rows / 200,000 test rows**, with **23 raw predictive features** once `id` and the target are removed; the positive rate is **0.30588**.

Data Quality is high: **no missing values, no duplicate rows**, and strong evidence the test set is an i.i.d. split of the same synthetic population — `id` is a contiguous row counter (train 0–299999, test 300000–499999), there is no date column, and category level sets are near-identical across splits (all low-cardinality columns identical; `nom_8` has 4 unseen test levels, `nom_9` 87 unseen out of 11,981). The strongest EDA signals were:

- **The ordinals’ alphabetical order $\approx$ their target-rate order** (correlations 0.994 / 0.994 / 0.971 for `ord_3/ord_4/ord_5`) — so alphabetical integer mapping is a valid ordinal encoding.

- **High-cardinality nominals are opaque 9-char hex hashes** (nom_5–nom_9, up to 11,981 levels) with no external semantics — signal lives entirely in the encoding, and external data is useless despite being permitted.
- **day/month are anonymized cycles** (1–7, 1–12) with no year anchor — no absolute timeline exists.

Annotation Guidelines. The label is `target` $\in \{0, 1\}$; submissions must be **probabilities/scores, not hard labels**, ranked by ROC-AUC — a rank metric, so calibration and thresholds are irrelevant by construction.

Feature Set. All 23 raw columns are modeled, in two representations: a **sparse one-hot matrix over train+test union categories (16,552 columns)** for the linear members, and an **integer-encoded frame** (ordinals mapped by their documented orders, plus sin/cos of day/month) for the GBDT members, with **fold-safe target encoding (smoothing 20, nested 4-subfold on train rows)** over nom_0–9 / ord_5 / day / month for the TE variants.

Group	Columns	Cardinalities
Binary (5)	<code>bin_0</code> – <code>bin_4</code> (<code>bin_3</code> T/F, <code>bin_4</code> Y/N $\rightarrow$ 0/1)	2 each
Nominal low-card (5)	<code>nom_0</code> – <code>nom_4</code>	3, 6, 6, 6, 4
Nominal high-card (5)	<code>nom_5</code> – <code>nom_9</code> (hex hashes)	222, 522, 1220, 2215, 11981
Ordinal (6)	<code>ord_0</code> – <code>ord_5</code> (documented orders)	3, 5, 6, 15, 26, 192
Cyclical (2)	<code>day</code> , <code>month</code>	7, 12

Splitting strategy. A single canonical **StratifiedKFold 5-fold CV (shuffled, seed = 42)** — the data is i.i.d. with no time or group structure, and stratifying on the 30.6% positive rate is cheap insurance. The dominant leakage risk here is **not the split but the encoders**: any target/likelihood encoding must be fit inside each training fold only (out-of-fold TE), never on the full training set — the discipline every TE member honors. The run itself was **leaderboard-blind** (OOF-only, closed-era benchmark run); the Public/Private figures quoted in Evaluation come from the study’s late-submission scoring, not from any in-run feedback.

Models & Architecture

Purpose of Architecture. A binary probability ranker that maximizes ROC-AUC. **Architecture Type** is unusual for this report set: the primary learner is **L2-regularized logistic regression on the sparse one-hot matrix** — on an all-categorical target with no continuous interactions, the linear model on OHE beats GBDT outright — with LightGBM members (fold-safe TE and native-categorical variants) kept for blend diversity. The champion is a **38-member kitchen-sink mega-blend** discovered at tree-search node #55, with **17 nonzero weights**.

Each linear member takes the same **Input Format** — the CSR one-hot design matrix — of **Input Dimension 16,552 sparse columns** per row; GBDT members consume the 23-column integer/TE frame (a 1-D vector; no spatial/sequence structure).

Architecture Description. The backbone member is LogisticRegression (`C` = 0.1, solver = lbfgs, `max_iter` = 2000, `tol` = 1e-5) on the full OHE matrix, solo OOF AUC 0.803103; the LRC lineage’s grid

refinement found the C optimum flat in [0.1, 0.15] with C = 0.12 best (0.803169). The blend layer is a convex weighted average of members' out-of-fold predictions, weights found by Dirichlet sampling (k = 800, cost-guard-coarsened to k = 200 on large blends) plus coordinate ascent. **Model Complexity** is dominated by the linear members (~16.5K weights each); the champion's weight mass is concentrated in the LR family — **lr_C0.12 at 0.3723**, lr_C0.1_root 0.1658, lr_C0.2 0.1403, with lgb_te_reg at 0.0732 and lr_C0.13 at 0.0447 — GBDT members contribute only a thin diversity sliver.

Training procedures

Training proceeds as a **staged ladder** on the shared StratifiedKFold split, each stage measured on the same OOF AUC scale:

Stage	Configuration	OOF AUC
linear solo	LR on full sparse OHE (16,552 cols), C = 0.1	0.803103
GBDT comparators	LGBM + fold-safe TE 0.797328 · LGBM native-categorical 0.776214	—
linear blend	greedy blend, LR C=0.1 (0.75) + C=0.2 (0.25)	0.803169
tree search harness_v3 (60/60 nodes, node #55)	38-member kitchen-sink mega-blend	0.803272

The tree search ran five lineages from the digit-verified root: **LRC** (C grid 0.07–0.5; best C = 0.12 → 0.803169), **LRPAIRS** (pairwise interaction crosses — **hurt everywhere**, best 0.796539 vs the 0.803103 base: explicit crosses damage a saturated sparse LR), **LGBTE** (tuning peaked at 0.798865 — the GBDT ceiling here is ~0.799), **SEEDBAG**, and **BLEND** (first ensemble seed beat the linear best at eval #10, 0.803226 > 0.803169). The champion came from the mandatory explore-burst mega-blend; both burst longshots passed the sanity gate (deep LGB 0.797445, near-unregularized LR C=1.0 on crosses 0.770255 — both far below, as predicted).

The **Loss Function** is log-loss (logistic regression; LightGBM binary objective), with model selection on OOF AUC. The **Optimization Algorithm** is L-BFGS for the linear members and histogram gradient boosting for the LGBM members — **not** SGD/ADAM — with blend weights optimized by Dirichlet search + coordinate ascent. The **Learning Rate** is *N/A for the champion's dominant members* (L-BFGS has no learning-rate hyperparameter); the LGBM members use boosting shrinkage 0.05 (pool) / 0.03 (regularized variant). There is **no Learning Rate Scheduler** (*N/A — no SGD-style optimizer in play*) and **no Batch Size** (*N/A — full-dataset solvers, not mini-batched*).

Training Duration for the full pipeline was ~75 min of wall-clock (headless run: EDA → prep → linear protocol → 60/60-node tree search); the root retrain took 12.5 s, and the cost guard coarsened every large blend evaluation (k = 800 → 200 whenever an eval exceeded the 45 s threshold; ~46–60 s observed on the mid-size blends, up to ~129 s on the 38-member mega-blends — always logged, never silent). **Training Memory** was not explicitly capped — a 300K × 16.5K sparse CSR matrix is modest on the single arm64 CPU machine, peak not recorded. **Transfer Learning** is *N/A (no pretrained weights)*; its analogue is **cross-competition experience injection** — the TASK-CAT-ONLY prior (“encoding is the whole game; linear-on-OHE competes with GBDT”) set the pool design, and the afsis lesson dictated reseeding the full linear pool

as cached-OOF first-generation nodes. **Data Augmentation** is *N/A (tabular)*; the analogues are the dual encoding representations and seed-bagging (LGB seeds 42/2024/777/4000–4004).

Reproducibility Standards are strict: fixed fold seed 42 with a persisted fold assignment (`folds.npy`), LightGBM determinism flags with 10 threads, and a **digit-verified root** — the tree search retrained the linear-stage best from scratch and matched the cached 0.803103 exactly before any search step. The honest ledger records `dedup_rejections = 10`, the full cost-guard and backtrack logs, and the stop reason (hard budget cap, 60/60 evaluated nodes). This was a native v3 run — the champion is the search’s own committed output.

Inference procedures

Decision Threshold is *N/A* — ROC-AUC is a rank metric, so we submit raw predicted probabilities and never threshold; calibration and any monotone post-processing are provable no-ops under AUC, so **no post-processing is applied**. The unseen test levels (4 in `nom_8`, 87 in `nom_9`) are handled by `handle_unknown="ignore"` in the OHE (all-zero row block) and the global-prior fallback in TE. **Inference Duration** was not separately profiled; scoring the 200,000 test rows is seconds per member on CPU, and **Inference Memory** is negligible (sparse dot products and tree traversal, not capped). The submission (`submission.csv` , 200K rows, `id/target`) was format-validated against `sample_submission.csv` .

Evaluation & Benchmarking

Task for Performance Evaluation. Rank the 200,000 test rows by probability of `target = 1`. **Performance Metrics.** ROC-AUC (maximize) is the sole competition metric; our champion reaches **CV OOF AUC 0.803272**, improving `0.803103 → 0.803169 → 0.803272` across the ladder, with the honest leave-fold-out weight re-fit at **0.803231** (in-sample weight optimism +0.000041). The tree search’s gain over the linear blend is +0.000103 — noise-level per-fold — so its real value here was **negative-result mapping**: interactions dead, GBDT ceiling `~0.799`, C optimum flat, confirming plain OHE+LR is essentially this feature space’s ceiling at this budget.

Performance Benchmarking. Per the set’s standing convention, **between-agent claims use the real leaderboard numbers; each agent’s local CV describes only its own pipeline’s internal view** (schemes differ across agents and are not comparable). The three-way comparison, from the study’s frozen score table:

Agent	Local CV AUC	Public LB	Private LB	Percentile
Our agent	0.803272	0.80801	0.80241	73.1
AIDE	0.803255	0.80790	0.80217	70.3
NVIDIA	0.77478	0.77641	0.77084	32.6

Our agent wins on both axes (local winner and leaderboard winner), placing **362/1341 (percentile 73.1)** on the final standings with Private 0.80241. The instructive structure is the two-tier result: our agent and AIDE independently converge on the same sparse-OHE + logistic-regression solution and finish within 0.00024 private AUC of each other — evidence that this really is the encoding-benchmark’s ceiling for a from-scratch

pipeline — while the NVIDIA reproduce-agent’s copied-kernel approach lands ~ 0.032 private AUC lower with a percentile of 32.6. On a competition built to test encoding judgment, deriving the right encoding from the data beat replaying someone else’s notebook by a wide margin, and the tiny CV $\rightarrow$ LB gap on our side (0.803272 local vs 0.80241 private) shows the StratifiedKFold estimate was honest.

Fields marked “not recorded”: training peak memory (not separately measured); inference duration and memory (not separately profiled). The run itself was leaderboard-blind (OOF-only); all LB figures come from the study’s frozen three-way score table.